Homework 2 Solutions

Shake It Off (The Budget Concerns)

Assessing the Economic Impact of Hosting Taylor Swift's Eras Tour

Quantitative Methods II — Winter 2025
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Acknowledgments

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3.1 Preliminaries

Q1: How many observations and variables are in the working dataset at this point?

A1: There are 14,416 observations and 42 variables in the working dataset. The data are at the county–industry level; that is, each row represents an industry in a specific county. There are 424 counties in the dataset.

Q2: According to the U.S. Department of Agriculture's Economic Research Service, the Rural-Urban Continuum Codes (RUCC) classify U.S. counties into nine categories based on metropolitan status, population size, and degree of urbanization. Code 1 corresponds to counties in metropolitan areas with a population of 1 million or more. How many such counties are represented in the working dataset?

A2: There are 424 counties from metropolitan areas with a population of 1 million or more.

Q3: If each row in the working dataset represents a county-level observation, and there are only a few hundred counties in U.S. metropolitan areas with populations of 1 million or more, why does the working data contain thousands of rows?

A3: Although it is at the county-level, there are multiple industry types. Hence, for each county, there will be N rows, each representing a different industry.

Q4: What do the variables 2001, 2002, up to 2023 represent in the dataset at this point?

A4: It represents the county-level GDP of a particular industry in a particular year.

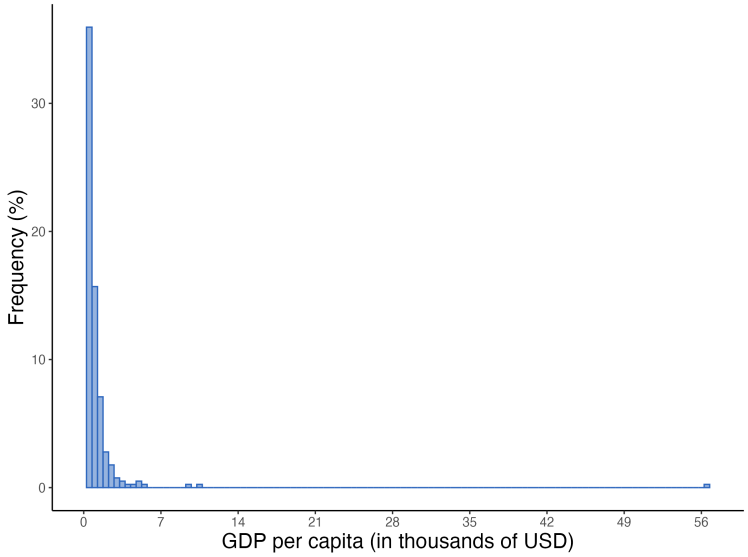
3.4.4 Transforming the Outcome Variables

Q5: In this section, we applied two transformations to the outcome variables: first, we converted GDP to per capita by dividing it by population, and second, we transformed the per capita values by taking their logarithm. **Q5a:** Why was it important to adjust for population size when analyzing GDP? **Q5b:** Based on the histograms you generated, what issue in the data prompted the need for a log transformation? **Q5c:** Did the log transformation resolve the issue you observed in the initial histogram of GDP per capita? Briefly explain your reasoning.

A5a: One expects GDP to increase as population increases: there is more consumption, more investment, more government expenditure, etc. Hence, GDP goes up. Without adjusting for population, you might mistakenly conclude that county i is more productive than county j simply because the former has a higher total GDP, when it may just have a larger population. GDP per capita shows how much production is generated per resident; it accounts for population size. Therefore, if county j has a smaller population but a higher GDP per capita than county i, it indicates stronger economic performance on a per-person basis, even if county i has a higher aggregate GDP.

A5b: Figure 1 shows a very right-skewed distribution. Extreme values can influence the overall findings of the regression analysis. To deal with this problem, we can use a log transformation.

Figure 1: Histogram of GDP per capita: Arts, Entertainment, and Recreation industry (2023)



Notes: This figure shows the distribution of GDP per capita (in thousands of USD) – considering only the Arts, Entertainment, and Recreation industry – across the metropolitan counties with a population of 1 million or more (2023). The histogram shows a very right-skewed distribution, where the median value is approximately $430. 24 counties do not have 2023 GDP per capita data for this industry. N = 395.

A5c: Figure 2 shows an improvement in the skewness of the distribution. Although it did not resolve the issue entirely, the distribution has a shorter right tail, which will help future regression analysis.

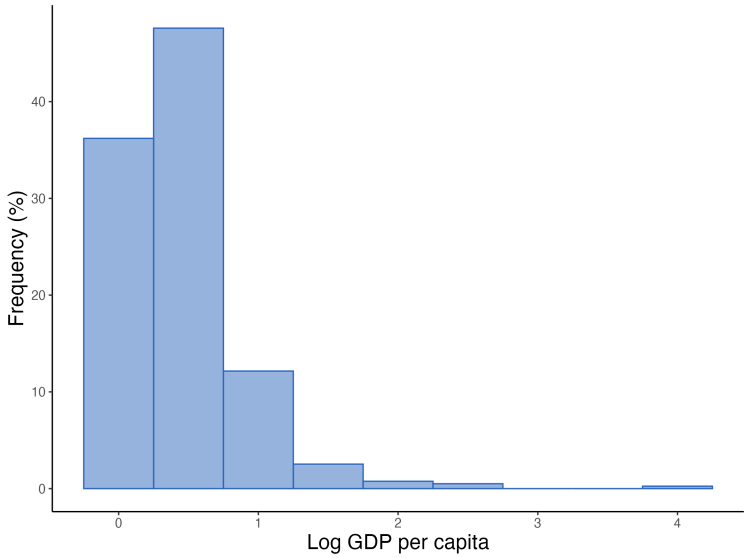
3.5 Analyzing Time Trends in Arts, Entertainment, and Recreation Industry (NAICS 71) Annual Log GDP per Capita

Q6: Examine the plot you generated. How does the trend in log GDP per capita for counties that hosted the Eras Tour compare to those that did not? Specifically, how do the slopes of these trends differ?

A6: Figure 3 shows the trends of the average of log-GDP per capita across metropolitan counties with a population of 1 million or more (2000-2023). The trends are split by counties that hosted the Eras Tour in 2023 (blue) and by those that did not (orange). The GDP per capita is specific to the Arts, Entertainment, and Recreation industry.

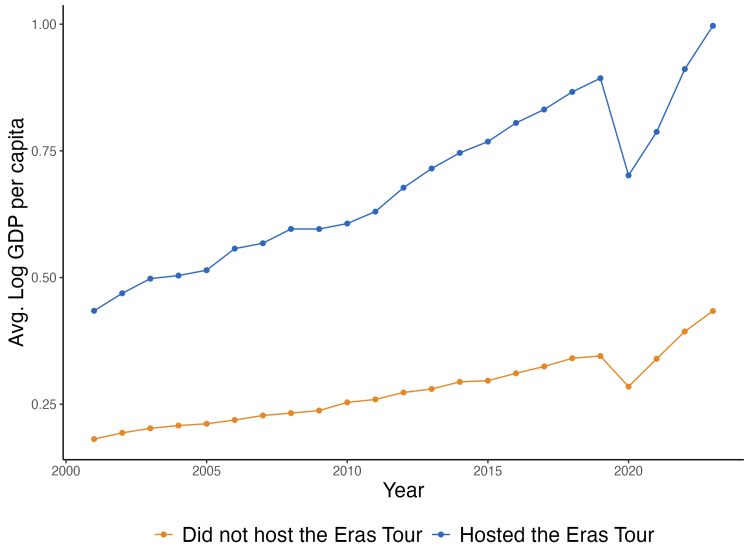
One sees that both groups experienced growth over the last 20 years, except for 2020, which was due to COVID (expected). The striking difference is the rate: counties that hosted the Eras Tour experienced, on average, higher GDP per capita growth than those that did not. In other words, the former grew at a faster pace than the latter. This indicates that the host group has, on average, a more dynamic economy than the non-host group, which is important to consider in future analysis.

Figure 2: Histogram of Log GDP per capita: Arts, Entertainment, and Recreation industry (2023)



Notes: This figure shows the distribution of the log of GDP per capita – considering only the Arts, Entertainment, and Recreation industry – across the metropolitan counties with a population of 1 million or more (2023). The log transformation addressed the variable's original skewness. 24 counties do not have 2023 GDP per capita data for this industry. N = 395.

Figure 3: Trends in GDP per capita, by hosting group (Arts, Entertainment, and Recreation industry (2000-2023))



Notes: This figure shows the average evolution of the log of GDP per capita – considering only the Arts, Entertainment, and Recreation industry – across the metropolitan counties with a population of 1 million or more (2000-2023). The trend is split by counties that hosted the Eras Tour (blue) and those that did not (orange). 24 counties do not have 2023 GDP per capita data for this industry. N = 395.

Q7: Why might this difference in pre-existing trends between the two groups create a problem when estimating the effect of hosting the Eras Tour on GDP?

A7: It creates a problem because there might be systematic differences between the two groups that could explain why counties that hosted the Eras Tour have had a higher GDP growth rate than those that did not. Maybe the first group has a more dynamic economy that attracts more investment, consumption, and pop-music events, such as the Eras Tour. In other words, there might be confounders (other variables) that help explain GDP growth, besides the event.

3.6 Visualizing Demographic Differences Across Counties that Hosted the Eras Tour and Those That Did Not

Q8: Examine the bar plots you generated. How do the demographic and socioeconomic characteristics of counties that hosted the Eras Tour compare to those that did not? Specifically, which characteristics show notable differences between the two groups?

A8: There are some notable differences between the two groups (see Figure 4 below). For example, the share of people with only a high school diploma is 25% higher in counties that did not host the Eras Tour than in those that did. On the other hand, the share of people with a bachelor's degree or higher is almost 27% higher in counties that hosted the Eras Tour compared to those that did not. The confidence intervals (black brackets) suggest that these differences are statistically significant. This finding also suggests that more highly educated people may live in counties that hosted the Eras Tour, which could positively impact GDP growth.

Another striking difference is the net migration. On average, host counties experienced negative net migration in 2023: more people left than entered. Maybe this is a result of the rising cost of living in more economically dynamic places. For those who did not host, the relation inverts.

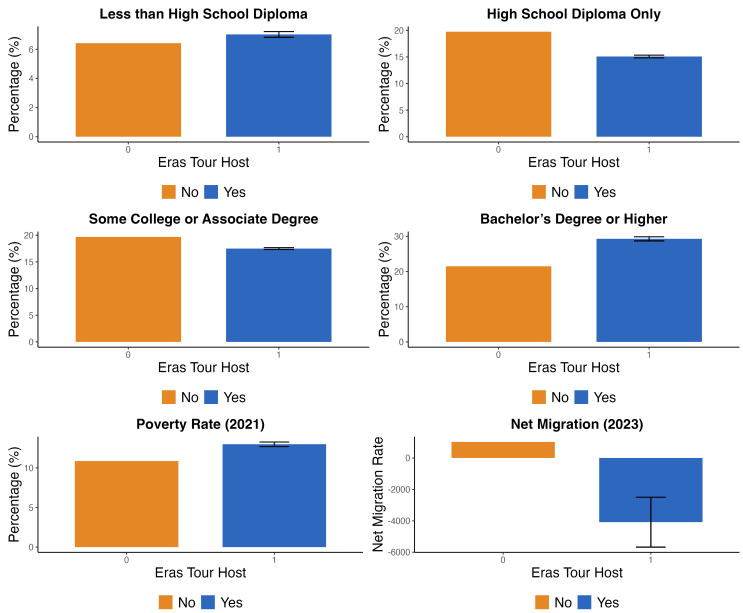
The share of people with less than a high school diploma and with some college or an associate degree, and the poverty rate are very similar across the groups, although the confidence intervals suggest a statistical difference in all measures.

All in all, the notable differences show higher-educated people in counties that hosted the Eras Tour. Also, negative net migration might indicate that people are moving from host counties to nearby ones with a lower cost of living.

Q9: Why might these differences in demographic and socioeconomic characteristics create a problem when estimating the effect of hosting the Eras Tour on GDP?

A9: These characteristics might be related both to the GDP and the fact that a particular county is hosting the tour. Maybe counties with more educated people attract venues more because of revenue expectations than places with fewer. Higher education also impacts the local GDP. Hence, ignoring these differences might bias the effect of hosting the Eras Tour on GDP, a phenomenon known as omitted variable bias.

Figure 4: Host vs. non-host the Eras Tour: demographics comparison panel



Notes: This figure shows a demographics comparison between counties that hosted the Eras Tour (blue) and counties that did not (orange). Education variables cover the 2018-2022 period. Poverty rate is from 2021, and net migration is from 2023. The analysis is at the county level. Black brackets represent 95% confidence interval. N = 424.

3.7 Multiple Regression Analyses

Q10: Look at the coefficient for Eras Tour Host in Model 5 under NAICS 71. What does the estimate tell you about how hosting the Eras Tour affects county-level GDP per capita in the Arts, Entertainment, and Recreation industry? Is the coefficient statistically significant?

A10: Column (model) 5 from Table 1 (see below) displays the regression coefficients with all the available controls. In terms of hosting the Eras Tour, counties that hosted the event in 2023 had, on average, an increase of approximately 1.6% in GDP per capita compared to counties that did not host the event. This increase concerns only the Arts, Entertainment, and Recreation industry. This coefficient is statistically significant at the 5% level.

Q11: How does controlling for GDP per capita in 2022 in Model 5 help address the issue raised in Q7 regarding pre-existing trends? Why is this step important in reducing omitted variable bias?

A11: If we go back to Figure 4, we see that host counties have had a greater growth rate compared to the ones that did not host the event. To account for this systematic difference, we control for the previous year's GDP per capita (2022). Controlling for that eliminated most of the apparent effect of hosting the Eras Tour on counties' GDP. At a 1% statistical significance, a 1% increase in the 2022 GDP results, on average, in a 1.038% increase in the 2023 GDP. Besides, the R^2 jumps from .43 to .99.

Table 1: Regression Results (Arts, Entertainment, and Recreation Industry)

	Dependent variable: Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	0.563***	0.328***	0.299***	0.297***	0.016**
	(0.092)	(0.076)	(0.076)	(0.077)	(0.008)
Share with Less than High School		3.891***	1.957*	1.850*	-0.051
		(0.787)	(1.054)	(1.118)	(0.117)
Share with High School Only		0.791*	0.578	0.547	-0.098*
		(0.461)	(0.464)	(0.476)	(0.050)
Share with Some College		1.875***	1.793***	1.757***	0.105
		(0.660)	(0.655)	(0.667)	(0.070)
Share with Bachelor's Degree or Higher		3.901***	3.826***	3.786***	0.079*
		(0.359)	(0.357)	(0.384)	(0.045)
Poverty Rate (2021)			0.017***	0.017***	0.002***
			(0.006)	(0.006)	(0.001)
Net Migration (2023)				-0.00000	0.00000
				(0.00000)	(0.00000)
Log GDP Per Capita (2022)					1.038***
					(0.006)
Constant	0.434***	-1.194***	-1.176***	-1.148***	-0.015
	(0.021)	(0.247)	(0.245)	(0.264)	(0.028)
Observations	395	395	395	395	391
R^2	0.087	0.415	0.426	0.426	0.994
Adjusted R^2	0.085	0.407	0.417	0.416	0.994
Residual Std. Error	0.400 (df = 393)	0.322 (df = 389)	0.319 (df = 388)	0.319 (df = 387)	0.033 (df = 382)
F Statistic	37.681*** (df = 1; 393)	55.135*** (df = 5; 389)	47.949*** (df = 6; 388)	41.014*** (df = 7; 387)	7,654.613*** (df = 8; 382)

Notes: This table shows the results from the multivariate regression estimation. The GDP value is related to the Arts, Entertainment, and Recreation industry. The dependent variable is the log of GDP per capita (2023). The independent variable of interest is an indicator for hosting the Eras Tour (1 = hosting; 0 = otherwise). The remaining independent variables are controls. *p<0.1; **p<0.05; ***p<0.01.

This means that the previous year's GDP has strong predictive power for the current year's GDP. Moreover, this suggests that these counties are likely to have higher GDP than the non-hosts, regardless of hosting the Eras Tour.

Q12: How do the changes in the coefficient estimates for eras tour host across the models address the issue raised in Q9 regarding omitted variable bias from demographic and socioeconomic differences?

A12: As we added more controls, especially for characteristics in which the two groups differ systematically, the estimated effect of hosting the Eras Tour diminished. This suggests that the coefficient estimate in Model 1 captured variation in GDP driven by pre-existing differences between host and non-host counties rather than by hosting the Eras Tour. In other words, the controls also have a role in the GDP's variation, and it is very likely that the estimate was suffering from omitted variable bias.

Q13: Examine the regression results for the other industries (NAICS 72, NAICS 54, and NAICS 11). Are the results consistent with the hypothesis that the Eras Tour had a greater economic impact on industries most likely to benefit from large events? Provide evidence from the models to support your response.

A13: Not entirely. On one hand, although we found a significant result for one industry most likely to be impacted, we did not find, after adding controls to our models, that Accommodation and Food Services, another industry in that group, was significantly affected.

On the other hand, it is true that industries less likely to be impacted did not benefit from hosting the concert at all. In fact, for the Agriculture, Forestry, Fishing, and Hunting industry, if we did not account for omitted variable bias, we could have misled policy makers, arguing that hosting the Eras Tour actually reduces the GDP, which is not true (see Figure 4). Overall, across all other industries, we find no statistically significant effect of hosting the event on GDP.

Table 2: Regression Results (Accommodation and Food Services Industry)

	Dependent variable: Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	0.518***	0.237***	0.187***	0.202***	0.006
	(0.094)	(0.072)	(0.070)	(0.070)	(0.006)
Share with Less than High School		4.577***	1.100	1.944*	-0.145
		(0.754)	(0.990)	(1.047)	(0.090)
Share with High School Only		-0.940**	-1.338***	-1.103**	-0.038
		(0.432)	(0.426)	(0.435)	(0.038)
Share with Some College		2.698***	2.530***	2.809***	-0.111**
		(0.623)	(0.604)	(0.612)	(0.054)
Share with Bachelor's Degree or Higher		3.684***	3.531***	3.840***	-0.0005
		(0.338)	(0.329)	(0.352)	(0.034)
Poverty Rate (2021)			0.029***	0.029***	0.0002
			(0.006)	(0.006)	(0.0005)
Net Migration (2023)				0.00001**	-0.00000
				(0.00000)	(0.00000)
Log GDP Per Capita (2022)					1.028***
					(0.005)
Constant	1.029***	-0.421*	-0.372	-0.591**	0.089***
	(0.021)	(0.233)	(0.226)	(0.243)	(0.021)
Observations	396	396	396	396	392
R^2	0.071	0.502	0.534	0.541	0.997
Adjusted R^2	0.069	0.495	0.527	0.532	0.997
Residual Std. Error	0.410 (df = 394)	0.302 (df = 390)	0.292 (df = 389)	0.291 (df = 388)	0.025 (df = 383)
F Statistic	30.266*** (df = 1; 394)	78.483*** (df = 5; 390)	74.273*** (df = 6; 389)	65.217*** (df = 7; 388)	14,035.170*** (df = 8; 383)

Notes: This table shows the results from the multivariate regression estimation. The GDP value is related to the Accommodation and Food Services industry. The dependent variable is the log of GDP per capita (2023). The independent variable of interest is an indicator of hosting the Eras Tour (1 = hosting; 0 = otherwise). The remaining independent variables are controls. *p<0.1; **p<0.05; ***p<0.01.

Table 3: Regression Results (Professional, Scientific, and Technical Services Industry)

	Dependent variable: Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	1.101***	0.506***	0.439***	0.452***	-0.0002
	(0.174)	(0.091)	(0.088)	(0.089)	(0.010)
Share with Less than High School		5.963***	1.332	2.107	-0.308**
		(0.988)	(1.284)	(1.364)	(0.147)
Share with High School Only		-0.688	-1.171**	-0.960*	-0.051
		(0.572)	(0.559)	(0.572)	(0.063)
Share with Some College		-0.998	-1.327*	-1.058	0.0001
		(0.814)	(0.787)	(0.801)	(0.087)
Share with Bachelor's Degree or Higher		7.669***	7.474***	7.759***	-0.017
		(0.440)	(0.425)	(0.458)	(0.067)
Poverty Rate (2021)			0.039***	0.039***	0.001
			(0.007)	(0.007)	(0.001)
Net Migration (2023)				0.00000*	0.00000
				(0.00000)	(0.00000)
Log GDP Per Capita (2022)					1.003***
					(0.006)
Constant	1.419***	-0.372	-0.301	-0.505	0.084**
	(0.041)	(0.306)	(0.295)	(0.319)	(0.035)
Observations	367	367	367	367	354
R^2	0.099	0.772	0.789	0.791	0.998
Adjusted R^2	0.096	0.769	0.786	0.787	0.998
Residual Std. Error	0.758 (df = 365)	0.383 (df = 361)	0.369 (df = 360)	0.368 (df = 359)	0.039 (df = 345)
F Statistic	39.916*** (df = 1; 365)	245.146*** (df = 5; 361)	224.880*** (df = 6; 360)	194.067*** (df = 7; 359)	18,369.470*** (df = 8; 345)

Notes: This table shows the results from the multivariate regression estimation. The GDP value is related to the Professional, Scientific, and Technical Services industry. The dependent variable is the log of GDP per capita (2023). The independent variable of interest is an indicator of hosting the Eras Tour (1 = hosting; 0 = otherwise). The remaining independent variables are controls. *p<0.1; **p<0.05; ***p<0.01.

Table 4: Regression results (Agriculture, Forestry, Fishing, and Hunting Industry)

	Dependent variable: Log GDP Per Capita (2023)				
	(1)	(2)	(3)	(4)	(5)
Eras Tour Host	-0.282***	-0.125	-0.098	-0.095	-0.016
	(0.103)	(0.098)	(0.098)	(0.099)	(0.032)
Share with Less than High School		0.238	2.024	2.164	-0.829*
		(0.973)	(1.300)	(1.371)	(0.451)
Share with High School Only		2.212***	2.381***	2.422***	-0.203
		(0.561)	(0.565)	(0.580)	(0.195)
Share with Some College		0.962	1.053	1.102	0.090
		(0.804)	(0.802)	(0.817)	(0.268)
Share with Bachelor's Degree or Higher		-0.439	-0.379	-0.323	-0.250
		(0.449)	(0.448)	(0.480)	(0.157)
Poverty Rate (2021)			-0.015**	-0.015**	0.004
			(0.007)	(0.007)	(0.002)
Net Migration (2023)				0.00000	-0.00000
				(0.00000)	(0.00000)
Log GDP Per Capita (2022)					0.884***
					(0.015)
Constant	0.326***	-0.222	-0.233	-0.272	0.110
	(0.022)	(0.304)	(0.303)	(0.326)	(0.107)
Observations	424	424	424	424	424
R^2	0.018	0.175	0.183	0.183	0.913
Adjusted R^2	0.015	0.165	0.171	0.170	0.911
Residual Std. Error	0.449 (df = 422)	0.414 (df = 418)	0.412 (df = 417)	0.412 (df = 416)	0.135 (df = 415)
F Statistic	7.535*** (df = 1; 422)	17.720*** (df = 5; 418)	15.589*** (df = 6; 417)	13.349*** (df = 7; 416)	543.417*** (df = 8; 415)

Notes: This table shows the results from the multivariate regression estimation. The GDP value is related to the Agriculture, Forestry, Fishing, and Hunting industry. The dependent variable is the log of GDP per capita (2023). The independent variable of interest is an indicator of hosting the Eras Tour (1 = hosting; 0 = otherwise). The remaining independent variables are controls. *p<0.1; **p<0.05; ***p<0.01.

Q14: Based on your regression analysis for the Arts, Entertainment, and Recreation industry (NAICS 71), calculate the implied increase in GDP for counties that hosted the Eras Tour. Use the provided hints and data to guide your calculations.

A14: To calculate the implied increase in GDP per capita for host counties, we must follow a few steps:

1. Convert the log effect to a proportional change and find the percentage increase:

$$((\epsilon^{0.016}) - 1) * 100 \approx 1.6\%$$

Hence, hosting the Eras Tour is associated with an average 1.6% increase in the county's GDP.

2. Calculate the estimated increase in thousand USD:

$$4,100,000 \times 1.6\% = 65,600$$

A 1.6% increase is, on average, approximately 66,000 thousands (or 66 million USD).

3. Calculate the per capita dollar impact

$$66,000,000 \div 2,300,000 \approx 28.7$$

Therefore, the estimated per capita economic boost is around $28.7.

Q15: Now, based on your calculation in Q14, what would you report to the mayor regarding the potential economic impact of hosting a Taylor Swift concert in San Diego?

A15:

To the Office of the Mayor of San Diego

From: Lucas I. Klotz, policy analyst

Subject: Should San Diego host Taylor Swift's the Eras Tour?

Many counties across the United States have reported positive economic impacts from hosting a Taylor Swift concert, part of the singer's Eras Tour. The event mobilizes venues, workforce, equipment, hotels and restaurants, and tourism to local cultural sites. This is good for the local economy: businesses increase their revenue, and governments collect more taxes. Does the Eras Tour actually provide this impact on the economy? Our findings show a potential positive impact on counties' GDP, but only for the Art, Entertainment, and Recreation industry. Another industry most likely to be impacted, Accommodation and Food Services, shows no impact from hosting the Eras Tour. Below, I discuss our results in more detail and the policy implications.

Our analysis provides a key insight: hosting the Eras Tour is associated with an increase in the Art, Entertainment, and Recreation industry's GDP. In particular, our regression analysis shows that

host counties increased their GDP per capita by an average of 1.6% in 2023 compared to counties that did not host Taylor Swift's tour. The result is statistically significant at the 5% level. The estimated economic benefit is an average increase of nearly $30 per citizen, or a total of around 66 million U.S. dollars.

Based on media reports, one might feel surprised and frustrated about the magnitude of this finding, as they expected a greater impact. We believe that many of these reports do not account for what we call the omitted variable bias. That is, ignoring other variables that might explain the change in GDP and incorrectly implying that it comes from a single reason; in this case, the Eras Tour. Indeed, when we account for the county's education, poverty, migration, and prior GDP performance, our analysis shows that each of these factors plays a role in the 2023 GDP, reducing the apparent effect of hosting Taylor Swift's concert. Thus, our comparison between host and non-host counties becomes more credible, as there are systematic differences between hosts and non-hosts. For example, in the last 20 years, host counties have had a faster GDP growth rate than non-hosts. This might suggest that these counties are likely to have higher GDP regardless of hosting the Eras Tour.

What about other industries? Did we find an impact? The short answer is no. Even for industries very likely to be affected by the Eras Tour, such as Accommodation and Food Services, after controlling for the variables mentioned above, we found no statistically significant effect of hosting the concert on the county's GDP. The same happened when looking at industries moderately likely to be impacted (Professional, Scientific, and Technical Services) and at the ones least likely (Agriculture, Forestry, Fishing, and Hunting). What do these findings tell us?

We believe that the Mayor's office should first account for all the costs involved in hosting Taylor Swift's the Eras Tour, like increasing police patrols, traffic, and overall urban cleaning. We say that because the event is associated with only a modest (at best) increase in the county's GDP, exclusively for a particular industry. If San Diego County designs (or has) a strategy to lower these costs, events like concerts are always welcome to increase the county's exposure to the national economy. If not, maybe we can find alternative ways to increase our region's economic performance.